Tomáš Kysela
ID: 503 177

1

$y' = 2e^{2x} e^{-y}$

$\frac{y'}{e^{-y}} = 2e^{2x}$

$\int \frac{dy}{e^{-y}} = \int 2e^{2x} dx$

$e^{y} = e^{2x} + C$

$\ln(e^{y}) = \ln(e^{2x} + c)$

$y = \ln(e^{2x} + c) \quad x \in \mathbb{R}$

$y(0) = \ln(2)$

$\ln(2) = \ln(e^{2 \cdot 0} + c)$

$\ln(2) = \ln(1 + c)$

$2 = 1 + c$

$c = 1$

$\boxed{y = \ln(e^{2x} + 1), x \in \mathbb{R}}$

②

$$y' = \frac{e^x - 1}{y}$$

$$0 = \frac{e^x - 1}{y} \rightarrow y = 0$$

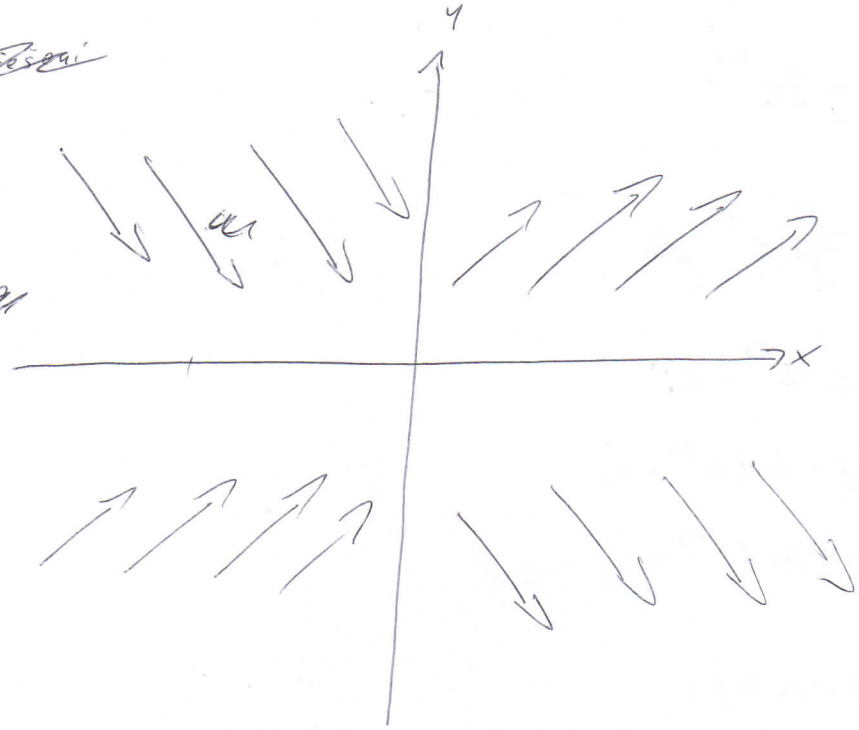
$$e^x - 1 = 0$$

$$e^x = 1$$

$$e^x = e^0$$

$$x = 0$$

n	s
c	d



a) $(-1; 1)$

$$y' = \frac{e^{-1} - 1}{1}$$

$$y' = \frac{1}{e} - 1 < 0 \ \ominus$$

b) $(1; 1)$

$$y' = \frac{e^1 - 1}{1}$$

$$y' = e - 1 > 0 \ \oplus$$

c) $(-1; -1)$

$$y' = \frac{e^{-1} - 1}{-1}$$

$$y' = 1 - e^{\frac{1}{}} > 0 \ \oplus$$

d) $(1; -1)$

$$y' = \frac{e^1 - 1}{-1}$$

$$y' = 1 - e < 0 \ \ominus$$

③

$y'' + 2y' - 3y = 0$

$\lambda^2 + 2\lambda - 3 = 0$

$(\lambda - 1)(\lambda + 3) = 0$

$\lambda = 1, -3$

$\pm 3 \pm 1$

$\lambda^2 + 2\lambda - 3 : (\lambda + 1) = \lambda$

$\lambda - 3$

$\lambda^2 + 2\lambda - 3 : (\lambda - 1) = \lambda + 3$

$3\lambda - 3$

OŘ:

$y(x) = ae^{x} + be^{-3x}$

$\lim_{x \to \infty} ae^{x} = \infty$

$\lim be^{-3x} = 0$

} fce v nekonečnu jde do nekonečna

$y'(x) = ae^{x} - 3be^{-3x}$

$y(0) = 1 \rightarrow 1 = ae^{0} + be^{-3 \cdot 0} \rightarrow 1 = a + b$

$y'(0) = -3 \rightarrow (-3) = ae^{0} - 3be^{-3 \cdot 0}$ $\quad (-3) = a - 3b$

$-4 = -4b$

$b = 1 \Rightarrow a = 0$

$\boxed{y(x) = e^{-3x}, \ x \in \mathbb{R}}$

(4)

$$g(x) = e^x + x \begin{cases} \nearrow e^x \xrightarrow{\text{zo kl.}} Ae^x \xrightarrow{\lambda=1} \text{korekce} \to xAe^x \\ \searrow x \to Bx + C \xrightarrow{\lambda=0} \text{bez korekce} \to Bx + C \end{cases}$$

$$\boxed{y_p = xAe^x + Bx + C}$$

$$\cancel{L = [xAe^x + Bx + C]'' + 2[xAe^x + Bx + C]' - 3[xAe^x + Bx + C] =}$$

$$L = [xAe^x + Bx + C]'' + 2[xAe^x + Bx + C]' - 3[xAe^x + Bx + C] =$$

$$[Ae^x(x+2)] + 2[Ae^x(x+1) + B] - 3[xAe^x + Bx + C] =$$

$$Ae^x + B(2-3x) - 3C$$